SYSDESIGN REFERENCE

API Gateway Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

Centralize edge concerns while keeping domain logic out of a potentially high-blast-radius tier.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / API-GATEWAY

SYSDESIGN

API Gateway Tradeoff Matrix

CORE IDEA

Centralize edge concerns while keeping domain logic out of a potentially high-blast-radius tier.

Reference architecture decision chain

01	02	03	04
Routing	Auth	Quotas	Transformation

Decisions to defend

+ Routing
+ auth
+ quotas
+ transformation
+ aggregation
+ and versioning.

Failure modes to test

+ Gateway bottlenecks
+ hidden coupling
+ policy drift
+ and cascading latency.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

API Gateway Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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